

Lecture 2 - Lexical Analysis - Introduction

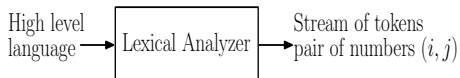
Compiler Design (CS 3007)

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Function of lexical analyzer



i : type of token
 j : number in the class of token

- Read source program, one character at a time
- Translate a sequence of characters to tokens (keywords, identifiers, constants, operators)
- Symbol table entry for new identifiers
- Forward tokens to parser
- Report errors like invalid identifier, undefined identifier, multiple declaration, invalid operators, keyword declared as identifier, etc

Function of lexical analyzer

Keywords
(token_type 1)

#1	if
#2	then
#3	else
#4	while
	:

Arithmetic
Operators
(token_type 2)

#1	+
#2	-
#3	*
#4	/
	:

Relational
Operators
(token_type 3)

#1	<
#2	>
#3	==
#4	!=
	:

Assignment
Operators
(token_type 4)

#1	=
#2	+ =
#3	- =
#4	* =
	:

Identifiers
(token_type 5)

#1	variable
#2	function
#3	array
#4	structure
	:

Token separators
(token_type 6)

#1	;
#2	blank
#3	,
#4	"
	:

if *a* < *b* **then**
a = *b* + *c*;

Lexical Analyzer

Stream of tokens
pair of numbers (*i*, *j*)

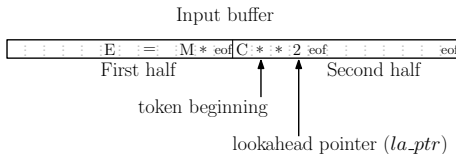
i: type of token
j: number in the class of token

Token	(<i>i</i> , <i>j</i>)
if	(1, 1)
blank	(6, 2)
<i>a</i>	(5, 1)
<	(3, 1)
<i>b</i>	(5, 1)
blank	(6, 2)
then	(1, 2)
=	(4, 1)
+	(2, 1)
<i>c</i>	(5, 1)
;	(6, 1)

Symbol Table

Name	Type	Number	...
#1 <i>a</i>	5	1	
#2 <i>b</i>	5	1	
#3 <i>c</i>	5	1	
:	:	:	:

Input Buffering



```

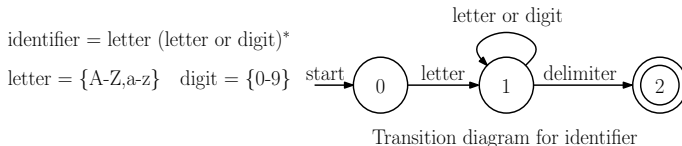
 $la\_ptr \leftarrow la\_ptr + 1;$ 
if  $la\_ptr = eof$  then begin
    if  $la\_ptr$  at the end of first half then begin
        reload second half;
         $la\_ptr \leftarrow la\_ptr + 1;$ 
    end
    else if  $la\_ptr$  at the end of second half then begin
        reload first half;
        move forward  $la\_ptr$  to the beginning of first half
    end
    else
        /*eof within a buffer signifying end of input*/
        terminate lexical analysis
    end if

```

Lookahead code with sentinels

- Lexical analyzer read characters from input buffer
- A pointer marks beginning of token being discovered
- *la_ptr* scans from beginning until the token is discovered

A simple lexical analyzer for identifier



```
state 0: C ← GETCHAR( );  
         if LETTER(C) then goto state 1;  
         else FAIL();  
state 1: C ← GETCHAR( );  
         if LETTER(C) or DIGIT(C) then goto state 1;  
         else if DELIMITER(C) then goto state 2;  
         else FAIL();  
state 2: C ← RETRACT( );  
         return (id, INSTALL( ));
```

Code for each state in the transition diagram

- GETCHAR() - advances the lookahead pointer and returns the next character
- LETTER(C) - returns **true** if C is a letter
- DIGIT(C) - returns **true** if C is a digit
- DELIMITER(C) - returns **true** if C is a character that can follow an identifier
- INSTALL() - returns a value that is a pointer to the symbol table
- FAIL() - returns error for invalid input
- RETRACT() - retracts delimiter from the identifier

Designing a lexical analyzer

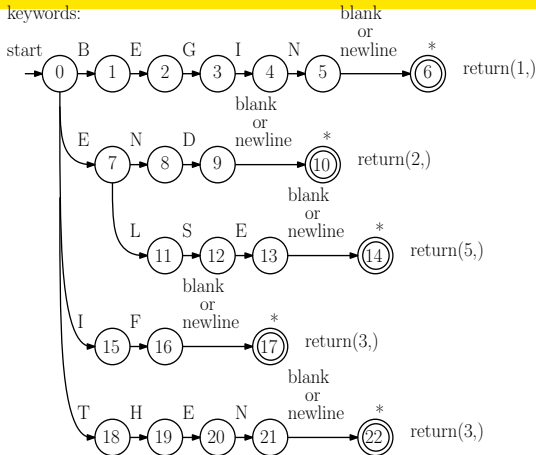
Token	Code	Value
begin	1	—
end	2	—
if	3	—
then	4	—
else	5	—
identifier	6	Pointer to symbol table
constant	7	Pointer to symbol table
<	8	1
<=	8	2
=	8	3
<>	8	4
>	8	5
>=	8	6

Tokens recognized

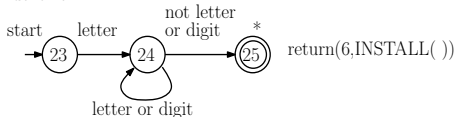
- Defining a token set
- Encoding each class of tokens
- Mapping values to tokens of same class

Transition diagrams for token recognizers

keywords:

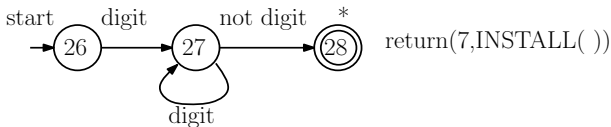


identifier:

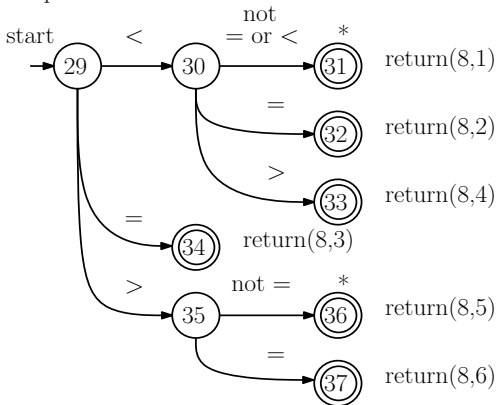


Transition diagrams for token recognizers

constant:



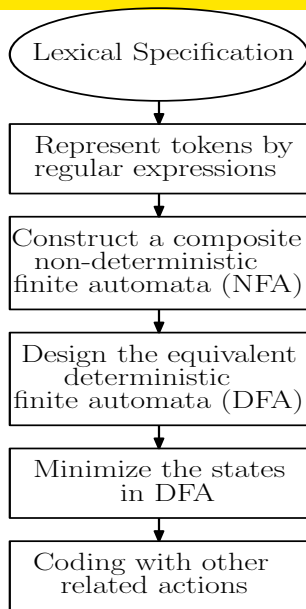
relops:



Combining the token recognizers

- Union of FSMs for identifiers, keywords, constants and relops
- A complex task
- Non-determinism may exist in the combined FSMs
- New transitions to be included
- Follow a proper procedure - simpler and error free
- Combine the token recognizers as an non-deterministic FSM
- Then obtain the equivalent deterministic FSM
- Minimize the states
- Add actions

Implementation of Lexical Analyzer

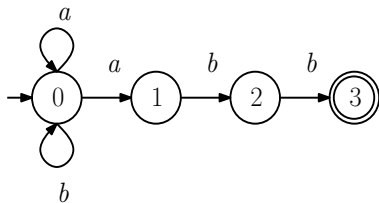


Lexical Specification and Regular Expressions

- An alphabet Σ is a finite set of symbols
- A string w is a finite sequence symbols, i.e. $w \in \Sigma^*$
- A language L is a set of strings over alphabet Σ , i.e. $L = \{w | w \in \Sigma^*\}$
- Regular expressions (REs) over alphabet Σ
 - ① ϵ is a RE denoting language $L = \{\epsilon\}$, where ϵ is an empty string.
 - ② For each $a \in \Sigma$, a is a RE denoting language $L = \{a\}$.
 - ③ If R and S are REs denoting languages L_R and L_S , respectively, then
 - Ⓐ $(R)|(S)$ is a RE denoting $L_R \cup L_S$.
 - Ⓑ $(R) \cdot (S)$ is a RE denoting $L_R \cdot L_S$.
 - Ⓒ $(R)^*$ is a RE denoting L_R^* .
- Example: following tokens are described by REs
 - keyword = BEGIN | END | IF | THEN | ELSE
 - identifier = letter (letter | digit)*
 - constant = digit⁺
 - relop = < | <= | = | <> | > | >=

Finite Automata

- A recognizer for a language L .
- Produces YES if an input string $w \in L$, otherwise NO.
- Consider a language $L = \{w \mid w \in \{a, b\}^* \wedge w \text{ ends with } abb\}$
- Regular expression for L is $R = R_1 \cdot R_2 = (a|b)^*abb$
- A simple recognizer for L is non-deterministic finite automata (NFA)



NFA for $(a|b)^*abb$

State	Input Symbol	
	a	b
0	$\{0, 1\}$	$\{0\}$
1	–	$\{2\}$
2	–	$\{3\}$

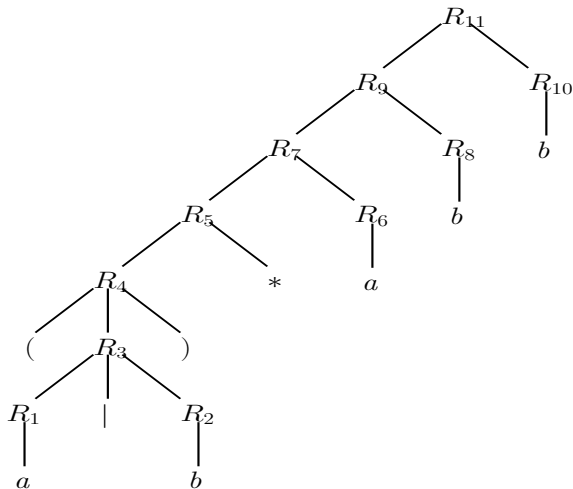
Transition table

State	Remaining Input
0	$aabb$
0	abb
1	bb
2	b
3	ϵ

Accepting $w = aabb$

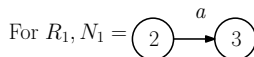
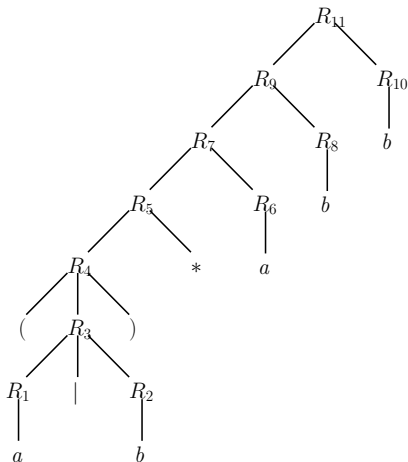
- Non-determinism cannot be implemented
- Convert NFA to deterministic finite automata (DFA)

Constructing NFA from Regular expression

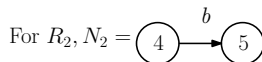
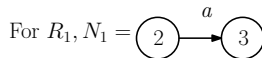
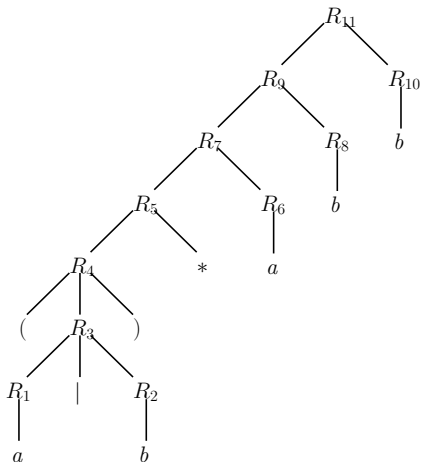


Decomposition tree for $(a|b)^*abb$

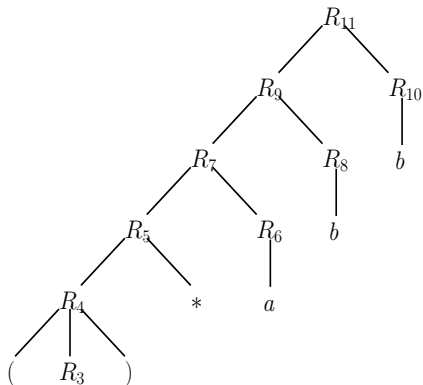
Constructing NFA from Regular expression



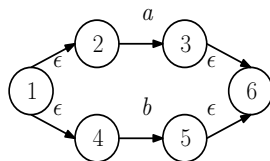
Constructing NFA from Regular expression



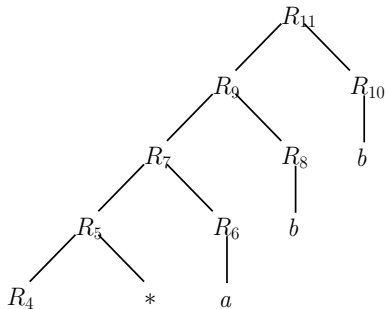
Constructing NFA from Regular expression



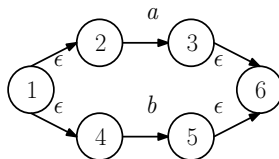
For $R_3 = R_1 | R_2$, $N_3 =$



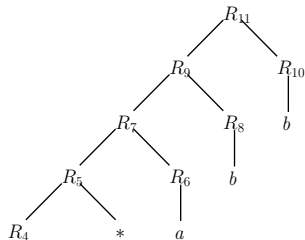
Constructing NFA from Regular expression



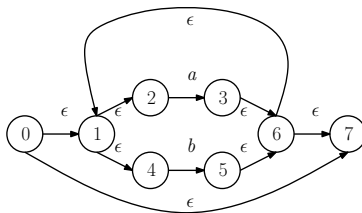
For $R_4 = (R_3)$, $N_4 = N_3 =$



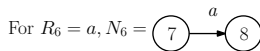
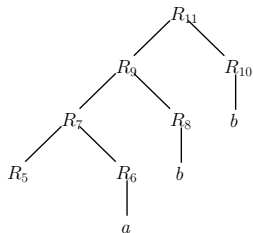
Constructing NFA from Regular expression



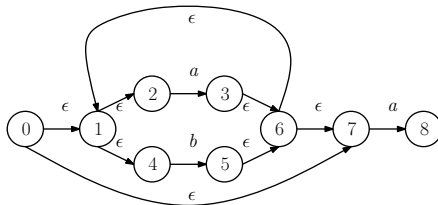
For $R_5 = R_4^*$, $N_5 =$



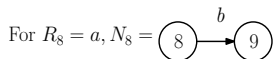
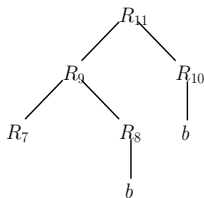
Constructing NFA from Regular expression



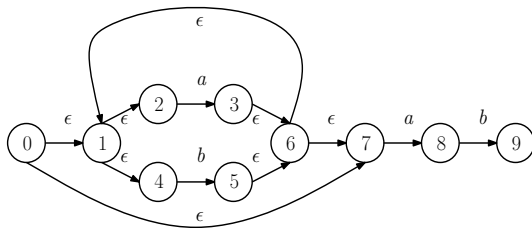
For $R_7 = R_5 \cdot R_6$, $N_6 =$



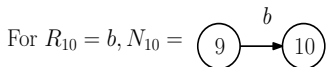
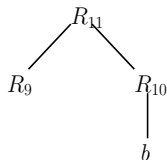
Constructing NFA from Regular expression



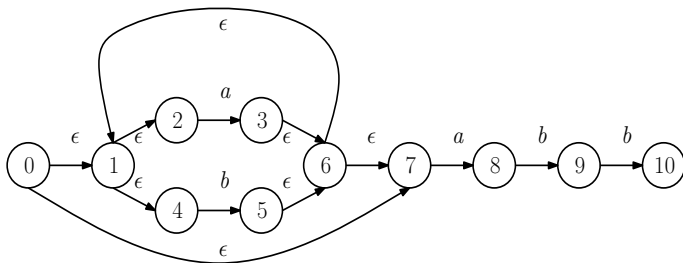
For $R_9 = R_7 \cdot R_8$, $N_9 =$



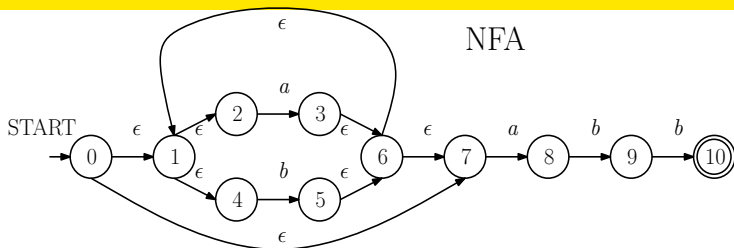
Constructing NFA from Regular expression



For $R_{11} = R_9 \cdot R_{10}$, $N_{11} =$



Constructing DFA from NFA



$$A = \epsilon - CLOSURE(\{0\}) = \{0, 1, 2, 4, 7\}$$

$$B = \delta(A, a) = \epsilon - CLOSURE(\{3\}) \cup \epsilon - CLOSURE(\{8\}) = \{1, 2, 3, 4, 6, 7, 8\}$$

$$C = \delta(A, b) = \epsilon - CLOSURE(\{5\}) = \{1, 2, 4, 5, 6, 7\}$$

$$\delta(B, a) = \epsilon - CLOSURE(\{3\}) \cup \epsilon - CLOSURE(\{8\}) = \{1, 2, 3, 4, 6, 7, 8\} = B$$

$$D = \delta(B, b) = \epsilon - CLOSURE(\{5\}) \cup \epsilon - CLOSURE(\{9\}) = \{1, 2, 4, 5, 6, 7, 9\}$$

$$\delta(C, a) = \epsilon - CLOSURE(\{3\}) \cup \epsilon - CLOSURE(\{8\}) = \{1, 2, 3, 4, 6, 7, 8\} = B$$

$$\delta(C, b) = \epsilon - CLOSURE(\{5\}) = \{1, 2, 4, 5, 6, 7\} = C$$

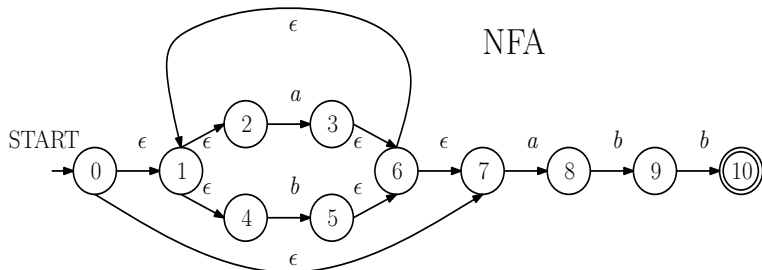
$$\delta(D, a) = \epsilon - CLOSURE(\{3\}) \cup \epsilon - CLOSURE(\{8\}) = \{1, 2, 3, 4, 6, 7, 8\} = B$$

$$E = \delta(D, b) = \epsilon - CLOSURE(\{5\}) \cup \epsilon - CLOSURE(\{9\}) = \{1, 2, 4, 5, 6, 7, 10\}$$

$$\delta(E, a) = \epsilon - CLOSURE(\{3\}) \cup \epsilon - CLOSURE(\{8\}) = B$$

$$\delta(E, b) = \epsilon - CLOSURE(\{5\}) = C$$

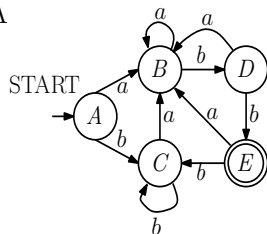
Constructing DFA from NFA



DFA

	NFA State	DFA State	Input	
			a	b
Start	{0, 1, 2, 4, 7}	A	B	C
	{1, 2, 3, 4, 6, 7, 8}	B	B	D
	{1, 2, 4, 5, 6, 7}	C	B	C
	{1, 2, 4, 5, 6, 7, 9}	D	B	E
Accept	{1, 2, 4, 5, 6, 7, 10}	E	B	C

Transition table



Transition diagram

Minimizing number of states in DFA

DFA

	State	Input	
		<i>a</i>	<i>b</i>
Start	<i>A</i>	<i>B</i>	<i>C</i>
	<i>B</i>	<i>B</i>	<i>D</i>
	<i>C</i>	<i>B</i>	<i>C</i>
	<i>D</i>	<i>B</i>	<i>E</i>
Accept	<i>E</i>	<i>B</i>	<i>C</i>

DFA with minimum number of states

	State	Input	
		<i>a</i>	<i>b</i>
Start	<i>A</i>	<i>B</i>	<i>A</i>
	<i>B</i>	<i>B</i>	<i>D</i>
	<i>D</i>	<i>B</i>	<i>E</i>
	<i>E</i>	<i>B</i>	<i>A</i>
Accept	<i>E</i>	<i>B</i>	<i>A</i>

$A = \{A, C\}$

Π_i : partition of set of states at step i

$\Pi_0 = \{\{A, B, C, D\}, \{E\}\}$, final and non-final states

$\delta(A, a) = \delta(B, a) = \delta(C, a) = \delta(D, a) = B \in \{A, B, C, D\}$

$\delta(A, b), \delta(B, b), \delta(C, b) \in \{A, B, C, D\}$ and $\delta(D, b) \in \{E\}$

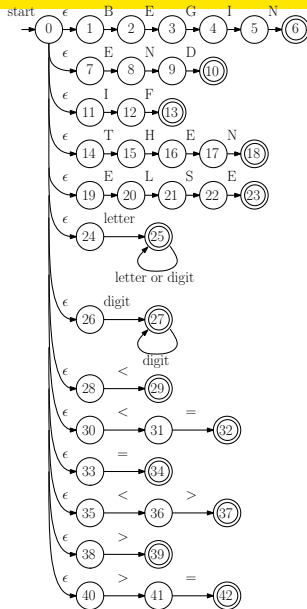
$\Pi_1 = \{\{A, B, C\}, \{D\}, \{E\}\}$

$\delta(A, a) = \delta(C, a) = B$ and $\delta(A, b) = \delta(C, b) = C$

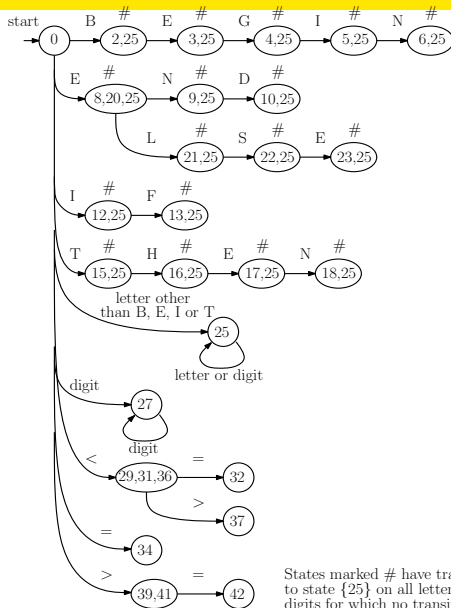
$\delta(B, a) = B$ and $\delta(B, b) = D$

$\Pi_2 = \{\{A, C\}, \{B\}, \{D\}, \{E\}\}$, final partition

Combined NFA for tokens



Combined DFA for tokens



Thank you